
Core bibliography and fact check of ergodicity notes

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Core bibliography and fact check of ergodicity notes

Compact core bibliography

Role	Core references	Why they belong
Foundations	[Bak14, Sil07]	Clear definitions of measure preservation, invariant sets, ergodicity, Birkhoff's theorem, recurrence, and stationary processes
Random dynamics	[Ara06]	Accessible introduction to random maps, cocycles, skew products, and stationary measures
Markov examples	[Kun20]	Concrete treatment of recurrence, stationary distributions, and empirical occupation measures
Nonstationary processes	[Gra80, Gar06]	Asymptotically mean stationary processes and cyclostationary or cycloergodic structure
Infinite ergodicity	[Aar97, Len18]	Standard framework, triviality of ordinary averages, and useful global observables
Multiplicative example	[Red90, Per96]	Typical-versus-average behavior and the Bernoulli-convolution stationary law
Applied time and ensemble averages	[Met14b, Gre19]	Weak ergodicity breaking, finite-time effects, aging, and apparent nonergodicity
Approximate ergodicity	[Fri11]	A clear account of epsilon-ergodicity and its difference from exact ergodicity

Useful formal supplements are [Pet02, Wal77, Dan08, Kac47, Boy83, Hur89, Lii06, Dom01, Mcc08b, Sch22].

Overall assessment

The notes have a strong pedagogical arc, but they currently move between four different ideas without always marking the transition:

- Exact measure-theoretic ergodicity of a measure-preserving system

- Equality of time and ensemble averages for a specified observable
- Stationarity or nonstationarity of a stochastic process
- Applied claims about ergodicity breaking from finite data

The article will be clearer if “ergodicity” is reserved for a dynamical or process-level property, while “mean ergodicity,” “ergodicity of an observable,” “epsilon-ergodicity,” and “apparent ergodicity breaking” are identified as weaker or operational notions.

Deterministic dynamical systems

Claim in the notes	Verdict	Required correction
Invariant sets satisfy $T(A) = A$.	Needs correction	Use $T^{-1}A = A$, or equality modulo null sets. Forward-image equality is not the general definition, especially when T is not invertible [Bak14].
Birkhoff averages equal the phase-space integral.	Correct with qualification	For $f \in L^1(\mu)$, the general limit is $\mathbb{E}_\mu[f \mid \mathcal{I}](x)$ almost everywhere. It equals $\int f d\mu$ almost everywhere only when the system is ergodic [Bak14].
The formula holds for every initial condition.	Incorrect	Replace “for every” by “for μ -almost every initial condition.” Exceptional initial conditions are allowed [Bak14].
Recurrence frequencies equal $\mu(A)$.	Correct with qualification	For each measurable A , the frequency equals $\mathbb{E}[1_A \mid \mathcal{I}](x)$ in general, and equals $\mu(A)$ almost everywhere in the ergodic case. Do not say “for almost all sets”; say “for every measurable set, for almost every initial condition.”
The mean return time is $1/\mu(A)$.	Correct under specific hypotheses	Define $\tau_A(x) = \inf\{n \geq 1 : T^n x \in A\}$. Kac’s formula uses the normalized conditional measure on A , and requires a probability-preserving ergodic setting with $\mu(A) > 0$: $\mathbb{E}_{\mu_A}[\tau_A] = 1/\mu(A)$ [Kac47].
Ergodic trajectories explore the full phase space.	Overstated	Ergodicity gives the correct frequency for every positive-measure measurable set, not necessarily visits to every point. Density of typical orbits requires additional topological assumptions such as full support and a countable basis of positive-measure open sets [Bak14].

Claim in the notes	Verdict	Required correction
A unique invariant probability measure must be ergodic.	Correct, but repair the proof	If a nontrivial invariant set exists, restrict the invariant probability measure to that set and normalize it. The proof should use $T^{-1}A = A$, not forward images, and should explicitly normalize $\mu(A \cap \cdot)$ by $\mu(A)$.

A corrected version of the central theorem is

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} f(T^n x) = \mathbb{E}_{\mu}[f \mid \mathcal{I}](x) \quad \text{for } \mu\text{-almost every } x.$$

If the system is ergodic and $\mu(\Omega) = 1$, the right-hand side is $\int_{\Omega} f d\mu$ almost everywhere [Bak14].

The discussion of Goldstein–Lebowitz typicality should be presented as a related foundational viewpoint, not as a consequence of the definition of ergodicity. Exact ergodicity and thermodynamic typicality are distinct claims; epsilon-ergodicity is a useful bridge between them [Fri11].

Stationary processes and path space

Claim in the notes	Verdict	Required correction
Any stochastic process can be represented by a deterministic shift on path space.	Correct with qualification	The canonical path-space construction is general, but the induced path measure is shift-invariant only when the process is stationary [Bak14].
The stationary distribution π_* induces the path-space measure.	Incorrect as written	The full joint law, or all finite-dimensional distributions, induces the path-space measure. The one-time marginal π_* alone is insufficient unless additional assumptions such as independence are supplied.
The path-space measure is $\pi_*^{\mathbb{Z}}$ for independent coordinates.	Correct with qualification	This is the iid case. For dependent processes, the path measure encodes all temporal correlations and finite-dimensional distributions.
Birkhoff gives an average of $f(T^n x_0)$ on path space.	Incorrect notation	Let $F(\omega) = f(\omega_0)$. Then $F(\theta^n \omega) = f(\omega_n)$, so the stochastic-process statement is $N^{-1} \sum_{n=0}^{N-1} f(X_n) \rightarrow \mathbb{E}_{\pi_*}[f]$ almost surely under shift ergodicity.
The result holds for every trajectory.	Incorrect	It holds for almost every path under the stationary path-space measure.

The distinction between the stationary distribution of a Markov kernel and the stationary path-space law should be made explicit. A kernel can have several stationary distributions; a specified stationary process has one specified path-space law. These are different uniqueness questions.

Skew-product representation

The skew-product idea is sound, but several formulas need repair.

For $\Omega = S \times \Xi^{\mathbb{Z}}$, use a consistent ordering such as

$$\Theta(x, \xi) = (F(x, \xi_1), \theta\xi),$$

not $(\theta\xi, F(x, \xi_1))$ if the state coordinate is listed first. The cocycle should also respect the shifted noise sequence:

$$\varphi(n, \xi)(x) = F_{\xi_n} \circ \dots \circ F_{\xi_1}(x), \quad \varphi(n+m, \xi) = \varphi(n, \theta^m \xi) \circ \varphi(m, \xi).$$

A broad random-map representation exists for Markov chains on standard Borel state spaces, but the inverse-CDF construction written in the notes is directly valid for real-valued states. For a general state space, write the transition kernel as $P(x, dy)$ and invoke a measurable randomization theorem. A finite-memory recursive process becomes Markov after augmenting the state, rather than necessarily being Markov in its original scalar coordinate.

The random-dynamical-systems framework is broader than Ito calculus: it can describe random forcing through a cocycle without invoking stochastic integration. It is safer to say that it includes many stochastic differential equations and random differential equations, rather than suggesting that it is an alternative formulation of every SDE without qualification [Ara06].

Discrete Ornstein–Uhlenbeck example

The stationary mean and variance are correct for

$$X_{n+1} = aX_n + \xi_{n+1}, \quad |a| < 1,$$

with iid Gaussian noise of mean μ_ξ and variance σ_ξ^2 :

$$m = \frac{\mu_\xi}{1-a}, \quad v = \frac{\sigma_\xi^2}{1-a^2}.$$

The displayed Gaussian density has a sign error. It should be

$$\pi_*^a(x) = \sqrt{\frac{1-a^2}{2\pi\sigma_\xi^2}} \exp \left[-\frac{1-a^2}{2\sigma_\xi^2} \left(x - \frac{\mu_\xi}{1-a} \right)^2 \right].$$

For zero-mean noise, the telescoping identity for X_n^2 is correct. The justification of the limiting terms needs stronger wording: X_N is not bounded or literally “of order one” almost surely. Instead, establish $X_N^2/N \rightarrow 0$ almost surely using the stationary finite-moment structure or a suitable moment and tail argument. The final statement should use a limit:

$$\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^{N-1} X_n^2 = \frac{\sigma_\xi^2}{1-a^2} \quad \text{almost surely}.$$

For symmetric Bernoulli noise, the support claim is cleanest for $0 < a < 1$:

$$\text{supp } (\pi_*^a) = \left[-\frac{1}{1-a}, \frac{1}{1-a} \right].$$

The stationary law is a measure, not generally a density. The Bernoulli-convolution facts should be stated as follows [Per96]:

- $0 < a < 1/2$: singular and supported on a Cantor-type set
- $a = 1/2$: uniform on $[-2, 2]$
- For Lebesgue-almost-every $a \in (1/2, 1)$: absolutely continuous with an L^2 density

It is too strong to say that every parameter in the exceptional measure-zero set is known to be singular. Singular parameters are known, including reciprocals of Pisot numbers, but the exceptional set should not be presented as completely classified [Per96]. The calculation at $a = 1/2$, $\mathbb{E}[X^2] = 4/3$, is correct.

Infinite ergodicity

The section has the right motivation but overstates several general conclusions.

- Replace “almost all sets are visited infinitely often” by “every positive-measure finite set is recurrent almost everywhere” in a conservative ergodic system. Arbitrary infinite-measure sets need not have the same return-time interpretation.
- The statement that integrable Birkhoff averages vanish almost everywhere is correct for conservative ergodic infinite-measure-preserving systems [Aar97, Len18].
- The expression $S_N f \sim a_N \zeta$ is not a universal theorem. Distributional limits and normalizing sequences depend on the system and observable. Use notation such as $S_N f / a_N \Rightarrow Z$ only after stating the hypotheses [Len18, Aki13].
- “Different trajectories retain order-one differences forever” is too broad. Persistent random amplitudes occur in particular distributional limit theorems, not in every infinite ergodic system or every observable.

For the simple symmetric random walk, counting measure is indeed an invariant infinite measure for the transition kernel. A walk started from a point is not stationary, but a sigma-finite stationary Markov measure can be constructed using counting measure. The local-time claim should not be written as $S_N f \sim c\sqrt{N}$ with an undefined random constant. Use a distributional statement such as

$$\frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} 1_{\{0\}}(X_n) \Rightarrow Z,$$

with the normalization and law of Z checked against a dedicated random-walk local-time reference. The parenthetical “half normal?” should be removed unless that reference is supplied.

The Pomeau–Manneville discussion needs a parameterization warning. Standard formulations use a piecewise map with a neutral fixed point, and different papers write the intermittency exponent in reciprocal conventions [Pom80, Bon17, Aki13]. The qualitative claims are sound: a power-law singularity near the neutral fixed point can produce a finite invariant measure below a threshold and an infinite invariant measure above it, with long return times and nontrivial normalized occupation sums. The exact map, density exponent, return-tail exponent, and threshold should be stated using one convention and supported by one source.

Nonstationary stochastic processes

The opening claim that nonstationarity “precludes ergodicity” should be removed. A nonstationary process does not generally define a shift-invariant path-space measure, so standard stationary-process ergodicity may be inapplicable rather than simply false. Also, nonstationarity does not by itself imply nonrecurrence. Asymptotically mean stationary processes provide a rigorous example in which strict stationarity fails but long-run averaged statistical behavior remains meaningful [Gra80].

The multiplicative random walk is a good pedagogical example, but the assumptions and indexing need correction:

$$X_{n+1} = z_n X_n, \quad X_n = X_0 \prod_{i=0}^{n-1} z_i.$$

To take logarithms, require $z_i > 0$ almost surely and $\mathbb{E} |\log z| < \infty$. Finite mean and variance of z alone do not guarantee this. Then

$$\frac{1}{n} \log \frac{X_n}{X_0} = \frac{1}{n} \sum_{i=0}^{n-1} \log z_i \rightarrow \mathbb{E}[\log z] \quad \text{almost surely}.$$

The comparison

$$\mathbb{E}[\log z] \leq \log \mathbb{E}[z]$$

is correct, with strict inequality for a non-degenerate positive random variable under the relevant integrability assumptions. The quantity $\lambda_n = \log X_{n+1} - \log X_n = \log z_n$ is an iid stationary observable of the noise shift, so the result is a law-of-large-numbers or Birkhoff statement for the transformed process—not for the original nonstationary variable X_n itself.

The claims about time averages being zero or infinite need qualification. If $\mathbb{E}[\log z] < 0$, then $X_n \rightarrow 0$ almost surely under the stated assumptions; if it is positive, typical paths grow exponentially. The boundary case and the behavior of finite-time averages require separate treatment. The ensemble mean is governed by $\log \mathbb{E}[z]$, which can differ from the typical growth rate, but the case $\mathbb{E}[z] = 1$ should not be described as either zero or infinite.

Practical considerations

The practical section should define the observable before using the word “ergodic.” Mean ergodicity usually concerns convergence of time averages of a specified observable to its ensemble mean in a mode such as mean square or probability. Agreement of means, moments, distributions, and finite-time estimators are different claims.

The following points are well supported:

- Finite data cannot generally establish an asymptotic property such as ergodicity [Loh24, Sch22].
- A mathematically ergodic process can look nonergodic over a finite observation window because relaxation is slow [Jeo12, Gre19].
- True weak ergodicity breaking can produce persistent trajectory-to-trajectory scatter and aging [Met14b, Bur10].

- Tests are observable- and model-dependent; there is no universal finite-data test that settles all forms of ergodicity [Dom01, Mag11, Jan15].
- Epsilon-ergodicity is a precise relaxation of exact ergodicity, but it is not identical to every applied use of “approximate ergodicity” [Fri11].

A useful practical distinction is therefore:

1. **Structural ergodicity:** a property of the full invariant system or stationary process.
2. **Observable ergodicity:** convergence for a chosen mean, moment, distribution, or time-averaged statistic.
3. **Finite-time representativeness:** whether the available trajectory is long enough for the desired estimator to be reliable.
4. **Operational approximation:** an explicit tolerance, confidence level, and trajectory-length criterion.

Highest-priority edits

Before circulating the article, the following corrections are essential:

- Replace forward-image invariance by pre-image invariance.
- Add “almost everywhere” to every Birkhoff-limit statement.
- Replace the phase-space integral by conditional expectation in the nonergodic case.
- Correct the Kac-return-time hypotheses and normalize the return-time measure.
- Correct the path-space and skew-product notation.
- Fix the sign in the Gaussian stationary density.
- Correct the Bernoulli-convolution statement for exceptional parameters.
- Replace universal claims about infinite-ergodic scaling by hypothesis-dependent distributional limits.
- Remove the claim that nonstationarity is automatically equivalent to nonergodicity.
- Add positivity and logarithmic-integrability assumptions to the multiplicative random walk.
- Define practical ergodicity in terms of a specified observable and finite-data criterion.

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